dune-rover executive brief

Executive brief: predicting slip and traversability of wheeled robots on Arabian dunes

Huda Mueen, `hudasol/dune-rover`, 2026-10-18. Generated by `scripts/build_report.py`; every number comes from a result file or an evidence row in `research/evidence.csv`. **This is a proposal and a research summary. No agreement, partnership or endorsement exists with any institution, and none is implied.** Heat and dust limits for people and electronics are the subject of the separate repository `hudasol/dhill` and are not repeated here. No cost is given.

Labels: **PUBLISHED** (read in a retrieved source), **SIMULATED BY ME** (output of this repository's code; simulation only, model assumption dependent, never field evidence), **PLANNED** (not run).

1. Question and Gulf relevance

How much does a small wheeled robot slip on a dune slope, can it be predicted, and how much of an open 30 m elevation map of UAE dunes can be trusted? No published slip against slope dataset was found for a wheeled robot on Arabian dune sand with the soil characterised; the only Arabian field item is a qualitative report of a rover getting stuck in loose sand in Oman (FLD-042). Open elevation data (Copernicus GLO-30, about 30 m) resolve the dunes but not their slip faces.

2. What is published (PUBLISHED; 171 rows, each traced to a source passage)

- **Earth dune slope curve.** At Dumont Dunes, California, a 340 kg test rover went from below 50 % slip at 12.5 degrees to about 75 % at 16 and about 90 % at 20 degrees; no soil parameters are reported (FLD-HV13-1, FLD-HV13-2).
- **Small rover on a sand bed.** A 13.5 kg rover traversing a 10 degree side slope of Toyoura sand showed a slip ratio of about 0.5 (TM-IS09-3).
- **Models need fitting per soil.** Bekker type drawbar errors of 24 to 27 % (median) on two lunar simulant datasets (SIM-025 to SIM-027); the Janosi shear modulus spans about two orders of magnitude between sources, so it is tuned, not transferred.

3. What is simulated by me (SIMULATED BY ME; simulation only, model assumption dependent)

A lumped slope to slip model with a published friction angle range and three assumed parameters, run over GLO-30 data for Liwa and the Rub' al Khali with Monte Carlo over the assumptions, a sub grid slip face detector validated against synthetic truth only, and a three class map (TRAVERSABLE, BLOCKED, UNKNOWN). The nominal climbing limit is 25.5 degrees; across the original assumption ranges it spans 14.9 to 31.6 degrees (5th to 95th percentile).

UNKNOWN share of cells by elevation model posting (original ranges; last column with the slip scale range widened on 2026-10-18):

Area	30 m	60 m	90 m	30 m, widened i0 range
A, south Liwa	8.6 %	9.1 %	7.1 %	33.9 %
B, Rub' al Khali megadunes	10.9 %	19.9 %	14.4 %	39.2 %
C, Rub' al Khali linear dunes (Saudi Arabia)	2.5 %	10.2 %	27.5 %	11.2 %

BLOCKED stays under 0.1 % at 30 m. When the resistance range is widened as well, UNKNOWN at 30 m reaches 92.6 % in A, largely because 9 to 12 % of those draws cannot move on level ground at all. Under the original ranges the map is robust to the slip parameters; under the widened ones it is not.

4. Negative results

- **The model predicts too little slip on both published anchors** (comparisons of unlike vehicles and soils). With the original ranges the nominal model crosses 50 % slip at 25.5 degrees against 12.6 degrees for the Dumont curve, and it gives 0.01 to 0.20 slip where about 0.5 was measured on Ishigami's 10 degree side slope.
- **Widening the slip scale range does not fix it.** The nominal model then crosses 50 % at 10.2 degrees, too early, and the envelope contains both anchors only because it is very wide; a post hoc fit still misses the shape of the Dumont curve (RMSE 0.071).
- **The planned pass or fail rule discriminates weakly.** On synthetic data with the revised slope bins and the amended entrapment rule, a wrong model with collapse above 20 degrees is called adequate in 52.7 % of replicates.
- Every hit and miss of the pre registered predictions is in `docs/PREDICTIONS.md`.

5. Error rate of the evidence table

Every row was re checked against its source by a separate agent with no access to the extraction notes. First check of the 170 rows as extracted: 9 material errors (5.3 %) and 51 with any discrepancy (30.0 %). Current table of 171 rows, latest verdict per row: 8 material (4.7 %). All are corrected. The checkers are language model agents; a human spot check by the author is planned.

6. The first field experiment (PLANNED)

A slope ladder with one small four wheel rover (5 to 25 kg) on one dune stoss slope: direct uphill runs in bins at 0, 5, 10, 12.5, 15, 17.5 and 20 degrees (25 degrees only if allowed; whether the 1 degree tolerance of these bins can be met on a real dune is unconfirmed, and the original bins at 5 degree spacing remain documented), a 12 run pilot, RTK ground truth, soil description at every station, calibration on runs up to 10 degrees and dated predictions committed before steeper runs (`docs/FIELD_VALIDATION_PLAN.md`).

7. The ask

A supervised field day on a dune site with a named supervisor and recovery support; the loan of one small wheeled rover; the loan of an RTK GNSS unit or a drone survey of the test slope; and written site access. Each has a fallback in `docs/PROPOSAL_EDGE.md`. Success means a complete dataset and a reported comparison with the committed predictions, even if the model fails.